

Cloud Migration Methodology and Playbook

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1 Introduction

Cloud migration is a strategic imperative for organizations seeking to modernize their technology estate, improve operational agility, reduce infrastructure costs, and accelerate innovation. Meridian & Associates has executed 250+ cloud migration programs, collectively migrating over 18,000 workloads to AWS, Microsoft Azure, and Google Cloud Platform. This playbook codifies our methodology, tools, and lessons learned into a repeatable framework that Meridian engagement teams can apply to migrations of any scale.

2 The 6R Migration Framework

Meridian's migration strategy is anchored in the 6R framework, which classifies every application and workload into one of six disposition categories. The 6R assessment is conducted during the Discovery phase and drives the migration approach, timeline, and cost for each workload.

2.1 Rehost (Lift and Shift)

Rehosting moves workloads to the cloud with minimal or no changes to the application architecture. This is the fastest migration path and is appropriate for applications with limited business differentiation, approaching end-of-life, or where the primary goal is data center exit. Rehosting typically achieves 20-30% cost savings from infrastructure consolidation and reserved instance pricing alone. Tools: AWS Application Migration Service, Azure Migrate, Google Migrate for Compute.

2.2 Replatform (Lift, Tinker, and Shift)

Replatforming involves targeted optimizations during migration without changing the core architecture. Common replatforming moves include migrating databases to managed services (RDS, Azure SQL, Cloud SQL), containerizing applications with Docker/Kubernetes, and replacing self-managed middleware with PaaS equivalents. Replatforming adds 15-25% effort over rehosting but typically delivers 35-50% cost savings through managed service efficiencies.

2.3 Refactor (Re-architect)

Refactoring involves significant architecture changes to leverage cloud-native capabilities such as serverless computing (Lambda, Azure Functions), event-driven architectures, microservices decomposition, and managed AI/ML services. Refactoring delivers the highest long-term value but requires the most investment (3-5x the effort of rehosting). Meridian recommends refactoring for strategic, high-value applications that will remain in the portfolio for 5+ years.

2.4 Repurchase (Drop and Shop)

Repurchasing replaces an existing application with a SaaS equivalent (e.g., on-premises Exchange to Microsoft 365, custom CRM to Salesforce, on-premises ITSM to ServiceNow). The migration effort shifts from infrastructure to data migration and process adaptation. Meridian's SaaS transition practice provides vendor evaluation, data migration, integration design, and change management for repurchase scenarios.

2.5 Retire

Retirement decommissions applications that are no longer needed, redundant, or superseded by other systems. Meridian's portfolio rationalization analysis typically identifies 10-20% of applications as retirement candidates, reducing

the migration scope and ongoing licensing and support costs. Retirement requires careful data archival and regulatory retention compliance.

2.6 Retain

Retention keeps applications in their current environment, either permanently (e.g., mainframe applications with prohibitive migration costs) or temporarily (deferred to a future migration wave). Retained applications require hybrid connectivity design (VPN, Direct Connect, ExpressRoute) and ongoing operational support planning.

3 Cloud Readiness Assessment

The Cloud Readiness Assessment (CRA) evaluates the organization's preparedness across six dimensions: application portfolio (technical debt, dependencies, licensing), infrastructure (network, compute, storage baseline), security and compliance (regulatory requirements, data residency, encryption standards), operations (monitoring, incident management, automation maturity), organizational readiness (cloud skills, team structure, vendor management), and financial readiness (FinOps capabilities, budget allocation, chargeback models).

Each dimension is scored on a five-level maturity scale: Initial (ad-hoc), Developing (documented), Defined (standardized), Managed (measured), and Optimizing (continuously improving). The CRA produces a roadmap of remediation activities that must be completed before or in parallel with migration waves.

4 Landing Zone Architecture

The landing zone is the foundational cloud environment that provides identity management, network architecture, security controls, logging, and governance guardrails for all migrated workloads. Meridian designs landing zones using the cloud provider's reference architectures as a starting point, customized for the client's specific requirements.

- AWS: Control Tower with Account Factory for Terraform (AFT), Service Control Policies, centralized logging via CloudTrail and Security Hub, transit gateway for network connectivity.
- Azure: Cloud Adoption Framework Landing Zones with Management Groups, Azure Policy for governance, Azure Monitor and Sentinel for observability and security, Hub-and-Spoke network topology.
- GCP: Resource hierarchy with Organizations, Folders, and Projects, Organization Policies, Cloud Logging and Security Command Center, Shared VPC for network isolation.

Landing zone deployment is automated using Infrastructure-as-Code (Terraform, Bicep, or CloudFormation) and version-controlled in Git. Meridian maintains landing zone blueprints for 8 industry verticals (Financial Services, Healthcare, Government, Retail, Manufacturing, Energy, Media, Technology) that incorporate industry-specific compliance requirements.

5 Migration Factory Model

For large-scale migrations (200+ workloads), Meridian operates a Migration Factory model that industrializes the migration process through standardized patterns, automation, and parallel execution. The factory model consists of three swim lanes operating concurrently.

- Assessment Lane: Application discovery, dependency mapping, 6R categorization, migration planning. Throughput: 50-100 applications per wave assessed in 2-3 weeks.
- Migration Lane: Execution of migration runbooks for each 6R pattern. Automated tooling for rehost and replatform.

Throughput: 50-100 workloads migrated per wave in 4-6 weeks.

- Validation Lane: Post-migration testing, performance benchmarking, security scanning, operational readiness. Throughput: aligned with Migration Lane output.

Each migration wave follows a 10-week cadence: 2 weeks assessment, 4-6 weeks migration, 2 weeks validation and optimization. Waves overlap by 4 weeks, enabling continuous migration throughput. A typical enterprise migration completes in 4-6 waves over 9-15 months.

6 Security and Compliance

Cloud security is integrated into every phase of the migration, not bolted on afterward. Meridian's cloud security framework addresses identity and access management (IAM), network security, data protection, workload protection, and security operations.

- FedRAMP: For federal government clients, Meridian ensures that all cloud services are FedRAMP authorized at the appropriate impact level (Low, Moderate, High). Our FedRAMP practice includes System Security Plan (SSP) development and continuous monitoring.
- HIPAA: Healthcare clients require BAA agreements, PHI encryption at rest and in transit, access logging, and breach notification procedures. Meridian maintains HIPAA-compliant landing zone templates for all three major cloud providers.
- PCI-DSS: Payment card processing environments require network segmentation, vulnerability management, access controls, and audit logging per PCI-DSS v4.0 requirements.
- SOC 2 Type II: Meridian assists clients in achieving SOC 2 compliance for cloud-hosted services, covering Trust Service Criteria for security, availability, processing integrity, confidentiality, and privacy.

7 FinOps and Cost Optimization

Cloud cost management is a critical capability that many organizations underestimate. Without disciplined FinOps practices, cloud costs can exceed on-premises costs within 18-24 months of migration. Meridian's FinOps framework establishes the people, processes, and tools needed for continuous cost optimization.

- Cost visibility: Tagging standards, cost allocation to business units, showback/chargeback models, dashboard and reporting (CloudHealth, Azure Cost Management, AWS Cost Explorer).
- Rightsizing: Continuous analysis of compute, storage, and database utilization to identify over-provisioned resources. Typical savings: 25-40% of compute spend.
- Reserved capacity: Commitment-based discounts (Reserved Instances, Savings Plans, Committed Use Discounts) for predictable workloads. Typical savings: 30-60% vs. on-demand.
- Spot/preemptible instances: For fault-tolerant, stateless workloads (batch processing, CI/CD, dev/test environments). Typical savings: 60-90% vs. on-demand.
- Waste elimination: Automated policies to shut down non-production resources outside business hours, delete unattached storage volumes, and remove unused elastic IPs and load balancers.

8 DevOps and SRE Enablement

Cloud migration provides an opportunity to modernize software delivery and operations practices. Meridian's DevOps enablement offering establishes CI/CD pipelines, infrastructure-as-code practices, automated testing, and observability platforms. Our SRE (Site Reliability Engineering) practice introduces service level objectives (SLOs), error budgets, toil

reduction, and incident management practices aligned with Google's SRE principles.

- CI/CD: GitHub Actions, Azure DevOps Pipelines, or GitLab CI. Automated build, test, security scan, and deployment for every code change.
- Infrastructure-as-Code: Terraform for multi-cloud, Bicep for Azure-only, CloudFormation for AWS-only. All infrastructure changes go through pull request review and automated validation.
- Observability: Datadog, Dynatrace, or cloud-native monitoring (CloudWatch, Azure Monitor, Cloud Monitoring) with distributed tracing, log aggregation, and alerting.
- Incident management: PagerDuty or Opsgenie for on-call rotation and escalation. Runbooks for common incidents. Blameless postmortems with published action items.

9 Engagement Approach

Cloud migration engagements are structured to deliver rapid, measurable progress through a phased wave model. The following outlines a typical engagement structure:

- Timeline: 3-4 months for Cloud Readiness Assessment and Landing Zone deployment; 9-15 months for full migration execution (4-6 waves); 3-6 months for post-migration optimization.
- Team Composition: 1 Engagement Partner, 1 Cloud Program Director, 1-2 Cloud Architects (certified in target platform), 2-4 Migration Engineers, 1 Security and Compliance Lead, 1 FinOps Analyst, 1 OCM Consultant (for change management and training).
- Key Deliverables: Cloud Readiness Assessment Report, 6R Application Disposition Matrix, Landing Zone Architecture Document, Migration Runbooks (per pattern), Wave Execution Reports, Post-Migration Validation Reports, FinOps Operating Model and Dashboard, Knowledge Transfer Package.

10 Related Case Studies & Key Personnel

The following case study and key personnel are directly relevant to cloud migration engagements:

- Public Sector Cloud Migration Case Study -- Federal agency FedRAMP-compliant migration of 400+ applications to AWS GovCloud; led by Dr. Priya Ramanathan, Lead Partner. Achieved \$12M annual infrastructure cost reduction.
- Jordan Lee, Senior Manager -- Cloud Architecture Lead, AWS Solutions Architect Professional and Azure Solutions Architect Expert certified. Specializes in large-scale migration factory operations and multi-cloud landing zone design.

11 Related Meridian Methodologies

Cloud migration engagements intersect with several other Meridian Advisory practice areas. The following documents provide complementary guidance:

- Cybersecurity Risk Assessment Methodology (v4.0) -- Security is integrated into every migration phase; the Cybersecurity Assessment framework is used for pre-migration security posture evaluation and post-migration validation.
- Standard Project Lifecycle Framework (v4.2) -- Cloud migrations follow the SPL governance structure, with migration waves mapping to SPL Build/Test/Deploy phases.
- Organizational Change Management Framework (v5.0) -- Cloud migrations require OCM support for IT operations teams transitioning from on-premises to cloud operating models.

- Data, Analytics & AI Strategy Guide (v1.0) -- Data platform modernization (data lake, lakehouse) is often executed as part of a cloud migration program.
- Enterprise ERP Implementation Strategy (v2.3) -- Cloud ERP migrations (e.g., on-premises SAP to S/4HANA Cloud) combine ERP transformation and cloud migration methodologies.